- Problem Statement ID –HC-05
- Problem Statement Title-CLIMATE AWARE EMERGENCY RESPONSE NETWORK
- Team Name-PHENOIX FORGE
- Team Leader Name- AKSHAT DUBEY
- Members Names-AKSHAT DUBEY,AMEYA SHROFF BASWARAJ BAWALGAVE,VISHAL HOLGIR

CLIMATE • TECHNOLOGY • RESILIENCE

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CLIMATE CHANGE & EMERGENCY RESPONSE NETWORK

Real-Time Data • Smarter Response • Safer Communities



ClimateAwareNetwork.org

A SAFER, GREENER FUTURE

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CliMATE

Climate-aware emergency response network: sense risk early, move help before disaster hits

THE PROBLEM

- Late, human-triggered warnings
- Siloed agencies and data
- Overlapping climate hazards
- Networks fail in a crisis

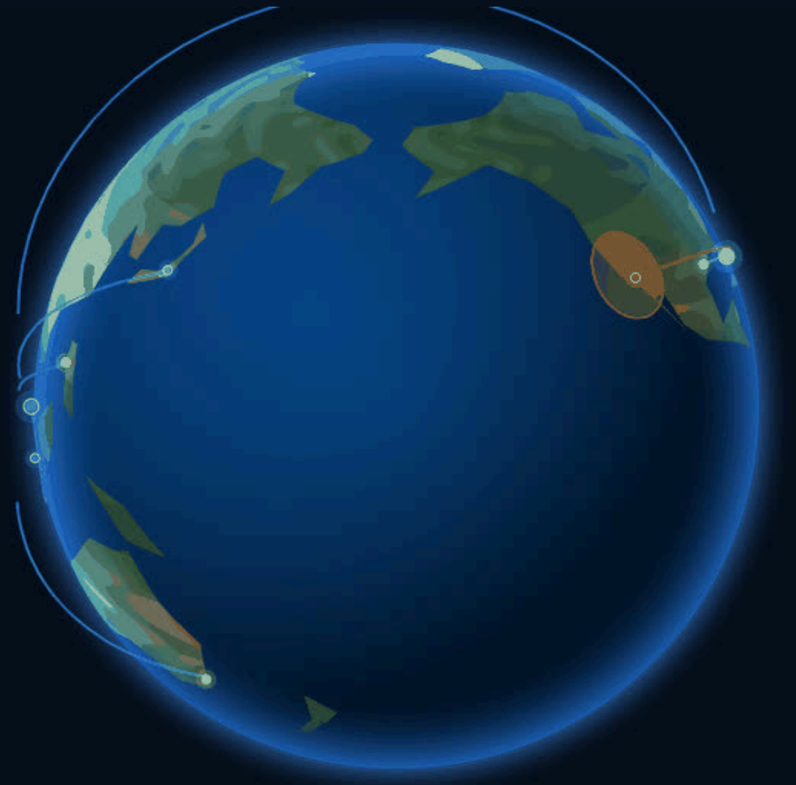
THE SOLUTION

Sense
Live sensors, satellites, weather feeds

Predict
Risk score and time to impact

Coordinate
Pre-position drones, supplies, care

Communicate
SMS, radio, mesh and satellite



LIVE 3D NETWORK MODEL

Hub Shelter Sensor Drone Hospital

Open interactive 3D version

WHAT MAKES IT UNIQUE

- Acts on forecast risk, not reports
- Self-rerouting network links
- Multi-hazard in one platform
- Mesh and satellite backup
- Vulnerability-first allocation

EXPECTED IMPACT

- Faster warnings and response
- Fewer people cut off
- Less supply waste
- Fairer protection for the vulnerable

CLIMATE CHANGE EMERGENCY RESPONSE NETWORK

Real-time data | Faster response | Safer communities

PROPOSED SOLUTION

A **community-centered** platform that connects residents, emergency responders, local authorities, and support organizations before, during, and after **climate-related emergencies**.

HOW IT WORKS

1. **Detect risks**
   Combine weather alerts, local hazard information, and community reports.
2. **Alert people**
   Send location-specific warnings and safety instructions through app, SMS, voice calls, or community radio.
3. **Connect people to help**
   Show nearby shelters, evacuation routes, cooling centers, medical services, and essential supplies.
4. **Coordinate response**
   Let residents report urgent needs while responders share incidents and available resources.
5. **Improve preparedness**
   Use feedback after each event to strengthen future response plans.

WHAT MAKES IT UNIQUE

- **Actionable alerts** with practical local next steps.
- **Two-way communication** so residents can report hazards and request help.
- A shared response **map** for risks, needs, and available resources.
- **Inclusive access** for people with limited internet or accessibility needs.
- **Community-informed** planning based on local knowledge.

One shared network. Faster response. Safer communities.

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TECHNICAL APPROACH

Technologies to be used

- AI and machine learning for hazard prediction and risk scoring
- HTML5 and CSS3 for the responder dashboard and alerts
- C++ for the fast network routing and simulation core
- SVG and CSS for overlays, legends and charts

Ai

HTML5

C++

CSS3

Methodology and process

- Flow: live hazard data (flood, fire, heat, storm) scores each zone; nearby nodes alert and routes reroute



Risk category: Low / Moderate / Elevated / High

Climate Change & Emergency Response Network

Connecting real-time climate alerts, emergency services, and communities.

Feasibility

Buildable with available resources

Simple Technology Stack
HTML, CSS, JavaScript, and map/weather APIs.

Available Data
Weather forecasts, hazard alerts, shelter locations, and emergency contacts.

Quick Prototype
Dashboard, alert feed, and emergency contact directory can be built during a hackathon.

Accessible Anywhere
Responsive web app for phones and computers.



Viability

Real impact, long-term value

Early Warnings
Sends timely alerts for floods, heatwaves, storms, and heavy rainfall.

Faster Coordination
Connects citizens, volunteers, hospitals, and rescue teams.

Safer Communities
Shares evacuation routes, nearby shelters, and emergency instructions.

Scalable Future
Can add live sensor data, AI-based risk prediction, and multilingual alerts.

THE RESULT

A connected, scalable emergency response network for climate-resilient communities.

Helps people prepare earlier, respond faster, and recover safer.

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Real-time data | Smarter response | Safer communities

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Impact

Tackling climate risks for a safer, greener future.

Reduced Climate Risks
Helps detect and predict extreme weather, floods, heatwaves and other climate-related hazards.

Faster Emergency Response
Real-time alerts and coordinated response help save lives and reduce damage.

Environmental Protection
Supports conservation, prevents deforestation and reduces carbon footprint.

Stronger Communities
Builds climate-resilient infrastructure and preparedness at local and regional levels.

Better Decision Making
Uses data, AI and technology for accurate forecasting and planning.

Early Warnings. Faster Actions. Greater Resilience.



Benefits

More safety. Less loss. A healthier planet.

Saves Lives
Timely alerts and early action reduce loss of life and injuries.

Reduces Economic Losses
Minimizes damage to homes, farms, businesses and infrastructure.

Improves Health & Well-being
Cleaner air, water and food security with less exposure to climate hazards.

Supports Sustainable Growth
Enables green jobs, clean energy and long-term development.

Builds a Resilient Future
A safer, healthier and more climate-adaptive world for future generations.

THE RESULT

A connected, scalable emergency response network for climate-resilient communities.

FUSION


HOSPITAL CAPE
CYCLONE // CAT 3
CYCLONE TRACK - 6HR FORECAST

Climate Change & Emergency Response Network

Real-time Data | Smarter Response | Safer Communities

Thank You!

Together for a
Safer, Greener Tomorrow

